How-To Library

Informatica

# Tuning Enterprise Data Catalog Performance in 10.5

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# Abstract

This article provides information about tuning Enterprise Data Catalog performance. Tuning Enterprise Data Catalog performance involves tuning parameters for metadata ingestion, ingestion database, search, and tuning data profiling. The performance of Enterprise Data Catalog depends on the size of data that needs to be processed.

# Supported Versions

- Enterprise Data Catalog 10.5

# Table of Contents

Overview. . . . 3
Minimum System Requirements for a Node in the Cluster. . . . 5
Enterprise Data Catalog Sizing Recommendations. . . . 6
Low. . . . 6
Medium. . . . 7
High. . . . 11
Very High. . . . 14
High Volume Sources. . . . 15
Metadata Extraction Scanner Memory Parameters. . . . 17
Tuning Configuration Parameters for Resources. . . . 29
Tuning Performance for Similarity. . . . 29
Enterprise Data Catalog Agent. . . . 30
Log Files. . . . 30
Performance Tuning Parameters for Profiling. . . . 31
Tuning Data Integration Service Parameters for Profiling. . . . 31
Data Integration Service Concurrency Parameters. . . . 35
Tuning Profiling Warehouse Parameters. . . . 37
Tuning Performance for Profiling on Spark Engine. . . . 38
Tuning Performance for Profiling on Blaze Engine. . . . 39
Memory Sizing to Run Profiles on Large Unstructured Files. . . . 40
Tuning Resource Configuration Parameters for Profiling. . . . 40
Profiling Recommendations for a Databricks Cluster. . . . 42
Catalog Export and Migration Recommendations. . . . 42
Catalog Re-index Estimates. . . . 44
Enterprise Data Catalog Performance Tuning Parameters. . . . 44
Performance Tuning Parameters for Data Asset Analytics. . . . 50
Tuning PostgreSQL Database Parameters for Data Asset Analytics. . . . 50
Tuning Oracle Database Parameters for Data Asset Analytics. . . . 51
Tuning Microsoft SQL Server Database Parameters for Data Asset Analytics. . . . 52
Tuning Microsoft Azure SQL Database Parameters for Data Asset Analytics. . . . 53

Tuning Amazon Web Services RDS for SQL Server Database Parameters. . . . . 54

# Overview

Performance sizing depends on the volume of data in the catalog, catalog concurrency in terms of concurrent user operations performed in the catalog, and resource concurrency.

You can extract metadata from external sources, such as databases, data warehouses, business glossaries, data integration resources, or business intelligence reports. The performance of Enterprise Data Catalog depends on the size of data being processed. Enterprise Data Catalog classifies data sizes into four categories based on the size of data.

The data sizes are classified into the following categories:

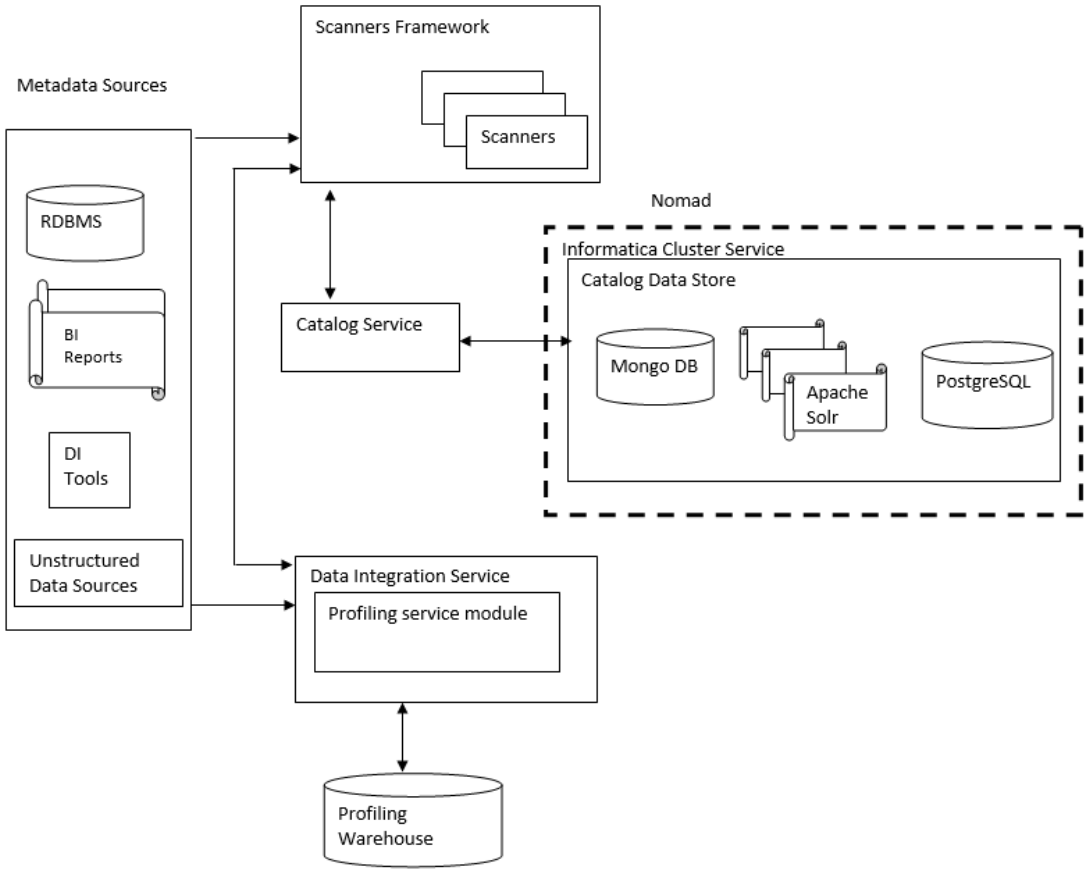
- Low
- Medium
- High
- Very High

Based on the data size, you can configure custom properties in Informatica Administrator to assign predefined parameter values for metadata ingestion, Mongo DB database tuning, and search. Alternatively, you can also individually configure the values for the performance tuning parameters based on your requirement. The profiling tuning involves tuning parameters for the Data Integration Service and the profiling warehouse database properties.

The metadata extraction, storage, search and related operations in an organization include the following phases:

1. Storing the extracted metadata into a catalog for ease of search and retrieval. A catalog represents a centralized repository for storing metadata extracted from different sources. This phase is referred to as the metadata ingestion phase. Enterprise Data Catalog uses Mongo DB as the database for ingesting data.
2. Validating the quality of data with data profiling that includes similarity profiling.
3. Searching for the related data assets in the catalog. Enterprise Data Catalog uses Apache Solr to search for data assets in the catalog.

The following image shows the high-level architecture of Enterprise Data Catalog:




The following table lists the components shown in the architecture diagram:

| Stage | Description |
|---|---|
| Scanner | A scanner is a plug-in component of Enterprise Data Catalog that extracts specific metadata from external data sources. |
| Data Integration Service | An application service that performs data integration tasks for Enterprise Data Catalog and external applications. |
| Profiling Warehouse | The data store where the profiling information is stored before the profiling information is moved to the catalog. |
| MongoDB | Enterprise Data Catalog data store for metadata and profiling results. |
| PostgreSQL | Enterprise Data Catalog data store for value frequency and similarity results. |
| Apache Solr | Index store used by Enterprise Data Catalog to store all the extracted metadata for search and retrieval. |
| Nomad | The orchestration service used by Enterprise Data Catalog. |

| Stage | Description |
|---|---|
| Catalog Service | The application service that runs Enterprise Data Catalog and manages the connections between service components and users. |
| Informatica Cluster Service | The Informatica Cluster Service runs and manages the services essential to run Enterprise Data Catalog. |

# Minimum System Requirements for a Node in the Cluster

The system requirements listed in the table are for a single node in the cluster. Make sure that you allocate the required additional system requirements for the operating system, processes, and other applications running on the node. The values listed are recommendations by Informatica for improving performance. You can increase the values based on your requirements.

## System Requirements

| System Requirement | Minimum Value |
|---|---|
| CPU | Eight physical cores with medium clock speed.<br>**Note:** For optimum performance, you must use cores with a minimum clock speed of 2.6 GHz. |
| Memory | 24 GB of unused memory available for use. |
| Hard disk drive | One high-capacity SSD hard disk drive with 200 GB to 1 TB capacity |
| File system | ext4 file system. |
| Network | Bonded Gigabit Ethernet or 10 Gigabit Ethernet between the nodes in the cluster. |
| **Note:** For high volume use cases, it is recommended that you have at least 32 GB of unused memory available for the MongoDB database. If the system does not have 32 GB of memory, you must configure 16 GB of swap space. | |

You must plan for the following additional system requirements per node based on the features that you plan to use:

## Advanced Scanners Server

Additional system requirements to configure the Advanced Scanners server to run as a separate Java Virtual Machine (JVM) on the node that hosts the Informatica domain:

| Number of CPU Cores | Memory | Free Disk Space |
|---|---|---|
| 2 | 4 GB | 20 GB |

### Advanced Scanners Resource

Additional system requirements to run an Advanced Scanners resource as a separate JVM on the node that hosts the Informatica domain. You need to increase the requirements appropriately based on the volume of metadata that the scanner extracts.

| Number of CPU Cores | Memory | Free Disk Space |
|---|---|---|
| 2 | 8-32 GB | A minimum of 50 GB |

### Similarity Discovery

Additional system requirements for the PostgreSQL database used as the data store for value frequency and similarity:

| Number of CPU Cores | Memory | Free Disk Space |
|---|---|---|
| 8 | 16 GB | A minimum of 200 GB for 2-4 resources that include approximately 10,000 columns in total.<br>You must increase the disk space based on the number of columns. |

# Enterprise Data Catalog Sizing Recommendations

Based on the size of data, you must add additional memory and CPU cores to tune the performance of Enterprise Data Catalog. You must also note the minimum number of cores that are required to deploy supported data sizes.

The size of data depends on the number of assets (objects) or the number of datastores. Assets include the sum total of databases, schemas, columns, data domains, reports, mappings and so on in the deployment environment. A datastore represents a repository that stores and manages all the data generated in an enterprise. Datastores include data from applications, files, and databases. Determine if you need to change the default data size for the Enterprise Data Catalog installation. Enterprise Data Catalog has low, medium, high, and very high data sizes or load types Data sizes are classified based on the amount of metadata to process and the number of nodes used to process metadata.

Performance sizing depends on the volume of data in the catalog, catalog concurrency in terms of concurrent user operations performed in the catalog, and resource concurrency.

**Note:** The minimum recommended load type for production deployment is medium. When the number of supported concurrent users for a load type increases, it is recommended that you move to the next higher load type.

Based on the parameters that affect performance sizing, you can plan to move to the next load type as required.

If you increase the number of nodes, you must perform the following steps:

1. Add the **Shared File Path** property in Informatica Cluster Service configuration.
2. Restart the Informatica Cluster Service.
3. Restart the Catalog Service.
4. Re-index the catalog.

## *Low*

Low represents two million assets or 30-40 datastores. The maximum user concurrency to access Enterprise Data Catalog is 20 users and the recommended resource concurrency is 1 medium data size resource or 2 low data size resources. It is not advisable to run high data size resources that require high memory allocation for low load type use cases.

For a low load type, it is recommended that you plan for one node to run the Informatica domain, one node for Enterprise Data Catalog, and an additional node to run similarity.

### System Requirements

The following table lists the recommended minimum system requirements for application services and infrastructure for low load type:

| Infrastructure Core Services | | | | | | Single Node Cluster for Services Associated with Informatica Cluster Service | Similarity Profiling Node |
|---|---|---|---|---|---|---|---|
| **Component** | **Infrastructure*** | **Data Integration Service+** | **Profiling Warehouse** | **Requirements for Advanced Scanners Server** | **Requirements for one Advanced Scanners Resource** | | |
| CPU Cores | 8 | 8 | 4 | 2 | 2 | 8 | 8 |
| Memory | 16 GB | 16 GB | 16 GB | 4 GB | 8-32 GB | 24 GB | 16 GB |
| Storage | 200 GB | 20 GB | 50 GB | 50 GB | 50 GB | 120 GB | 200 GB |

*Includes Informatica domain services, Model Repository Service, Catalog Service, Content Management Service, Advanced Scanners Service, Similarity profiling data store database, and Informatica Cluster Service.

+The hardware sizing requirement listed is for a use case that includes profiling.
**Note:** You can run the Data Integration Service on the same node as the domain or on a separate node. If the Data Integration Service used for profiling co-exists on the Infrastructure node, the total hardware sizing is the sum of Data Integration Service and Infrastructure hardware components.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the CPU requirements listed in the table by 1.5 to 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 5 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

**Note:** The Catalog Service heap size is 2 GB. The Maximum Heap Size for Data Integration Service is 2 GB.

**Service Level Agreement**

The Service Level Agreement (SLA) to process metadata extraction and profiling for a low data size is approximately two business days. The SLA calculation is for a basic RDBMS resource such as Oracle.

**Calculation**

The complexity of transformations present in the extracted metadata affect the memory required for lineage. Ensure that you increase the memory for infrastructure as required.

## *Medium*

Medium represents 20 million assets or 200-400 datastores. The maximum user concurrency to access Enterprise Data Catalog is 50 and the recommended resource concurrency is 3 medium data size resources or 1 high data size resource that requires high memory allocation. A medium data size resource requires a minimum of three nodes to run the cluster.

For a medium load type, it is recommended that you plan for one node to run the Informatica domain, three nodes to run Enterprise Data Catalog and an additional node to run similarity. You can configure the three nodes on which you run Enterprise Data Catalog as data nodes and processing nodes.

You can choose to use the minimum system requirements or system requirements for a high volume source as necessary. The system requirement tables list the components and values for infrastructure, Data Integration Service, and profiling warehouse.

## Minimum System Requirements

The following table lists the recommended minimum system requirements for application services and infrastructure for medium load type:

| **Infrastructure Core Services** | | | | | | **Informatica Cluster Service and Associated Services** | | **Similarity Discovery** |
|---|---|---|---|---|---|---|---|---|
| **Component** | **Infrastructure*** | **Data Integration Service†** | **Profiling Warehouse** | **Requirements for Advanced Scanners Server** | **Requirements for one Advanced Scanners Resource** | **Requirements per Node** | **Total Requirement for Three Nodes** | **Requirements for the Node** |
| CPU Cores | 24 | 32 | 8 | 4 | 2 | 8 | 24 | 8 |
| Memory | 32 GB | 64 GB | 32 GB | 4 GB | 8-32 GB | 24 GB | 72 GB | 16 GB |
| Storage | 200 GB | 100 GB | 200 GB | 50 GB | 100 GB | 600 GB | 1.8 TB | 500 GB |

* Includes Informatica domain services, Model Repository Service, Catalog Service, Content Management Service, Advanced Scanners Service, and Informatica Cluster Service.

†The hardware sizing requirement listed is for a use case that includes profiling.
**Note:** You can run the Data Integration Service on the same node as the domain or on a separate node. If the Data Integration Service used for profiling co-exists on the Infrastructure node, the total hardware sizing is the sum of Data Integration Service and Infrastructure hardware components.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the CPU requirements listed in the table by 1.5 to 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 20 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

**Note:** The Catalog Service heap size is 4 GB. The Maximum Heap Size for Data Integration Service is 10 GB.

**Service Level Agreement**

The following table describes the Service Level Agreement (SLA) to process metadata extraction and profiling for medium load type:

| Process | SLA |
|---|---|
| Metadata extraction | Approximately one business week to accumulate 20 million assets based on metadata source types.<br>The SLA calculation is for a basic RDBMS resource such as Oracle or PowerCenter. |
| Profiling | Approximately one business week in addition to metadata extraction SLA.<br>The SLA calculation is for a basic RDBMS resource, such as Oracle, with 10K sampling rows for profiling and 10 data domains.<br>**Note:** The time taken to complete profiling is linearly proportional to the total number of data domains selected for profiling. |

**Calculation**

When you calculate the system requirements, make sure that you add the system requirements for infrastructure and three cluster nodes, and as required for profiling and Catalog Service tuning.

## Recommended System Requirements for High Volume Sources

A high volume source might include a single large resource or large number of resources. As a general guideline, if the object count in a resource exceeds the recommended limit for medium scanner type, the resource is considered to be a high volume source. For more information about the object count, see the "Metadata Extraction Scanner Memory Parameters" on page 17 topic.

**System Requirements**

The following table lists the system requirements for application services and infrastructure for high volume sources for the medium load type:

| Infrastructure Core Services | | | | | | Informatica Cluster Service and Associated Services | | Similarity Discovery |
|---|---|---|---|---|---|---|---|---|
| Component | Infrastructure* | Data Integration Service+ | Profiling Warehouse | Requirements for Advanced Scanners Server | Requirements for one Advanced Scanners Resource | Requirements per Node | Total Requirement for three nodes | Requirements for the Node |
| CPU Cores | 24 | 64 | 8 | 4 | 2 | 12 | 36 | 8 |
| Memory | 32 GB | 128 GB | 32 GB | 4 GB | 8-32 GB | 48 GB | 144 GB | 16 GB |

| Infrastructure Core Services | | | | | | Informatica Cluster Service and Associated Services | | Similarity Discovery |
|---|---|---|---|---|---|---|---|---|
| Component | Infrastructure* | Data Integration Service+ | Profiling Warehouse | Requirements for Advanced Scanners Server | Requirements for one Advanced Scanners Resource | Requirements per Node | Total Requirement for three nodes | Requirements for the Node |
| Storage | 200 GB | 100 GB | 200 GB | 50 GB | 100 GB | 600 GB | 1.8 TB | 1 TB |

* Includes Informatica domain services, Model Repository Service, Catalog Service, Content Management Service, Advanced Scanners Service, and Informatica Cluster Service. You can configure one Data Integration Service or configure Data Integration Service on two nodes where each node includes 32 CPU cores and 64 GB of memory.

+The hardware sizing requirement listed is for a use case that includes profiling.
**Note:** You can run the Data Integration Service on the same node as the domain or on a separate node. If the Data Integration Service used for profiling co-exists on the Infrastructure node, the total hardware sizing is the sum of Data Integration Service and Infrastructure hardware components.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the CPU requirements listed in the table by 1.5 to 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 20 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

For more information about high volume sources, see the "High Volume Sources" on page 15 topic.

**Service Level Agreement**

The following table describes the SLA to process metadata extraction and profiling for medium load type for high volume sources:

| Process | SLA |
|---|---|
| Metadata extraction | Approximately three business days to accumulate 20 million assets based on metadata source types.<br>The SLA calculation is for a basic RDBMS resource such as Oracle or PowerCenter. |
| Profiling | Approximately four business days in addition to metadata extraction SLA.<br>The SLA calculation is for a basic RDBMS resource, such as Oracle, with 10K sampling rows for profiling and 10 data domains.<br>**Note:** The time taken to complete profiling is linearly proportional to the total number of data domains selected for profiling. |

The throughput or performance depends on the number of columns in the table, file size, file type, complexity of mapping, size of the reports, file content, and number of data domains for profiling.

## *High*

High represents 50 million assets or 500-1000 data stores. The maximum user concurrency to access Enterprise Data Catalog is 75 and the recommended resource concurrency is 5 medium data size resources or 1 high data size resource that requires high memory allocation. A high data size resource is recommended to be run on 6 nodes.

You can choose to use the minimum system requirements or system requirements for a high volume source as necessary. The system requirement tables list the components and values for infrastructure, Data Integration Service, and profiling warehouse.

### Minimum System Requirements for Existing High Load Type Configuration Users

As an existing user, you must plan for an additional node to run similarity. If the Informatica Cluster Service runs on a cluster with six nodes and you plan to upgrade to 10.5, see the following table for the minimum system requirements:

| **Infrastructure Core Services** | | | | | | **Informatica Cluster Service and Associated Services** | | **Similarity Discovery** |
|---|---|---|---|---|---|---|---|---|
| **Component** | **Infrastructure*** | **Data Integration Service+** | **Profiling Warehouse** | **Requirements for Advanced Scanners Server** | **Requirements for one Advanced Scanners Resource** | **Requirements per Node** | **Total Requirement for six nodes** | **Requirements for the Node** |
| CPU Cores | 48 | 32 | 16 | 8 | 2 | 8 | 48 | 8 |
| Memory | 64 GB | 64 GB | 64 GB | 4 GB | 8-32 GB | 24 GB | 144 GB | 16 GB |
| Storage | 300 GB | 500 GB | 500 GB | 50 GB | 100 GB | 1 TB | 6 TB | 1 TB |

* Includes Informatica domain services, Model Repository Service, Catalog Service, Content Management Service, Advanced Scanners Service, and Informatica Cluster Service. You can configure one Data Integration Service or configure Data Integration Service on two nodes where each node includes 32 CPU cores and 64 GB of memory.

+The hardware sizing requirement listed is for a use case that includes profiling.
**Note:** You can run the Data Integration Service on the same node as the domain or on a separate node. If the Data Integration Service used for profiling co-exists on the Infrastructure node, the total hardware sizing is the sum of Data Integration Service and Infrastructure hardware components.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the CPU requirements listed in the table by 1.5 to 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 100 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

**Note:** The Catalog Service heap size is 9 GB. The Maximum Heap Size for Data Integration Service is 20 GB.

**Service Level Agreement**

The following table describes the SLA to process metadata extraction and profiling for high load type:

| Process | SLA |
|---|---|
| Metadata extraction | Approximately two business weeks to accumulate 50 million assets based on metadata source types.<br>The SLA calculation is for a basic RDBMS resource such as Oracle or PowerCenter. |
| Profiling | Approximately two business weeks in addition to metadata extraction SLA.<br>The SLA calculation is for a basic RDBMS resource, such as Oracle, with 10K sampling rows for profiling and 10 data domains.<br>**Note:** The time taken to complete profiling is linearly proportional to the total number of data domains selected for profiling. |

**Calculation**

When you calculate the system requirements, make sure that you add the system requirements for infrastructure and six cluster nodes, and as required for profiling and Catalog Service tuning. For better throughput with additional hardware capacity, see the performance tuning section for each component in this article.

## Recommended System Requirements for New Users and Existing Medium Load Type Configuration Users Planning to Move to a Higher Load Type

If you are a new user or an existing user planning to move from an existing load type to a higher load type, see the listed system requirements.

**System Requirements**

The table lists the recommended system requirements for application services and infrastructure for the following scenarios:

- You are a new user with Informatica Cluster Service running on a three-node cluster with the load type configured as high.
- You are an existing user with Informatica Cluster Service running on a three-node cluster with the load type configured as medium. You plan to upgrade to version 10.5 and configure a higher load type.

,

| Infrastructure Core Services | | | | | | Informatica Cluster Service and Associated Services | | Similarity Discovery |
|---|---|---|---|---|---|---|---|---|
| Component | Infrastructure* | Data Integration Service[1] | Profiling Warehouse | Requirements for Advanced Scanners Server | Requirements for one Advanced Scanners Resource | Requirements per Node | Total Requirement for three nodes | Requirements for the Node |
| CPU Cores | 48 | 32 | 16 | 8 | 2 | 16 | 48 | 8 |
| Memory | 64 GB | 64 GB | 64 GB | 4 GB | 8-32 GB | 48 GB | 144 GB | 16 GB |

| Infrastructure Core Services | | | | | | Informatica Cluster Service and Associated Services | | Similarity Discovery |
|---|---|---|---|---|---|---|---|---|
| Component | Infrastructure* | Data Integration Service[1] | Profiling Warehouse | Requirements for Advanced Scanners Server | Requirements for one Advanced Scanners Resource | Requirements per Node | Total Requirement for three nodes | Requirements for the Node |
| Storage | 300 GB | 500 GB | 500 GB | 50 GB | 100 GB | 2 TB | 6 TB | 1 TB |

* Includes Informatica domain services, Model Repository Service, Catalog Service, Content Management Service, Advanced Scanners Service, and Informatica Cluster Service. You can configure one Data Integration Service or configure Data Integration Service on two nodes where each node includes 32 CPU cores and 64 GB of memory.

[1] Indicates that you can use the specified memory for Data Integration Service, or use two Data Integration Service nodes of 32 cores and 64 GB each. The hardware sizing requirement listed is for a use case that includes profiling.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the CPU requirements listed in the table by 1.5 to 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 100 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

You can configure all the three service nodes as data nodes and processing nodes.

For more information about high volume sources, see the "High Volume Sources" on page 15 topic.

**Service Level Agreement**

The following table describes the SLA to process metadata extraction and profiling for high volume sources for the high load type:

| Process | SLA |
|---|---|
| Metadata extraction | Approximately one-two business weeks to accumulate 50 million assets based on metadata source types.<br>The SLA calculation is for a basic RDBMS resource such as Oracle or PowerCenter. |
| Profiling | Approximately two business weeks in addition to metadata extraction SLA.<br>The SLA calculation is for a basic RDBMS resource, such as Oracle, with 10K sampling rows for profiling and 10 data domains.<br>**Note:** The time taken to complete profiling is linearly proportional to the total number of data domains selected for profiling. |

The throughput or performance depends on the number of columns in the table, file size, file type, complexity of mapping, size of the reports, file content, and number of data domains for profiling.

**Recommendations**

You can consider the following recommendation when you want to optimize performance for high volume sources in the high load type object limit:

- To increase the profiling throughput and use the recommended hardware effectively, see the "setConcurrencyParameters Script" on page 36 topic to tune the parameters as necessary.

## *Very High*

Very high represents 100 million assets or 500-1000 data stores with high volume sources. The maximum user concurrency to access Enterprise Data Catalog is 100 and the recommended resource concurrency is 5 medium data size resources or 1 high data size resource that requires high memory allocation. A very high data size resource is recommended to be run on 6 nodes.

For a very high load type, it is recommended to plan for one node for the Informatica domain, six nodes to run Enterprise Data Catalog, and an additional node to run similarity. Out of the six nodes on which you run Enterprise Data Catalog, you can configure three nodes as data nodes and three nodes as processing nodes. Apache Solr can run on all six nodes.

The system requirement tables list the components and values for infrastructure, Data Integration Service, and profiling warehouse.

### Minimum System Requirements

The following table lists the recommended minimum system requirements for application services and infrastructure:

| Infrastructure Core Services | | | | | | Informatica Cluster Service and Associated Services | | Similarity Discovery |
|---|---|---|---|---|---|---|---|---|
| Component | Infrastructure* | Data Integration Service+ | Profiling Warehouse | Requirements for Advanced Scanners Server | Requirements for one Advanced Scanners Resource | Requirements per Node | Total Requirement for Six Nodes | Requirements for the Node |
| CPU Cores | 48 | 64 | 32 | 8 | 2 | 16 | 96 | 8 |
| Memory | 64 GB | 128 GB | 64 GB | 4 GB | 8-32 GB | 48 GB | 288 GB | 16 GB |

| Infrastructure Core Services | | | | | | Informatica Cluster Service and Associated Services | | Similarity Discovery |
|---|---|---|---|---|---|---|---|---|
| Component | Infrastructure* | Data Integration Service+ | Profiling Warehouse | Requirements for Advanced Scanners Server | Requirements for one Advanced Scanners Resource | Requirements per Node | Total Requirement for Six Nodes | Requirements for the Node |
| Storage | 300 GB | 500 GB | 500 GB | 50 GB | 100 GB | 2 TB | 12 TB | 2 TB |

* Includes Informatica domain services, Model Repository Service, Catalog Service, Content Management Service, Advanced Scanners Service, and Informatica Cluster Service. You can configure one Data Integration Service or configure Data Integration Service on two nodes where each node includes 32 CPU cores and 64 GB of memory.

+The hardware sizing requirement listed is for a use case that includes profiling.
**Note:** You can run the Data Integration Service on the same node as the domain or on a separate node. If the Data Integration Service used for profiling co-exists on the Infrastructure node, the total hardware sizing is the sum of Data Integration Service and Infrastructure hardware components.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the CPU requirements listed in the table by 1.5 to 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 200 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

**Note:** The Catalog Service heap size is 18 GB. The Maximum Heap Size for Data Integration Service is 20 GB.

**Service Level Agreement**

The following table describes the SLA to process metadata extraction and profiling for high load type:

| Process | SLA |
|---|---|
| Metadata extraction | Approximately two-four business weeks to accumulate 100 million assets based on metadata source types.<br>The SLA calculation is for a basic RDBMS resource such as Oracle or PowerCenter. |
| Profiling | Approximately two business weeks in addition to metadata extraction SLA.<br>The SLA calculation is for a basic RDBMS resource, such as Oracle, with 10K sampling rows for profiling and 10 data domains.<br>**Note:** The time taken to complete profiling is linearly proportional to the total number of data domains selected for profiling. |

**Calculation**

When you calculate the system requirements, make sure that you add the system requirements for infrastructure and six cluster nodes, and as required for profiling and Catalog Service tuning. For better throughput with additional hardware capacity, see the performance tuning section for each component in this article.

## *High Volume Sources*

A high volume source might include a single large resource or large number of resources. As a general guideline, if the object count in a resource exceeds the recommended limit for medium scanner type, the resource is considered to be a

high volume source. For more information about the object count, see the "Metadata Extraction Scanner Memory Parameters" on page 17 topic.

### Recommendations to Optimize the Performance

You can tune the components to optimize the performance when you scan a high volume source, or when one or more of the following use cases is true:

- Re-indexing a large catalog.
- Re-assigning the connection assignment parameters.
- Deleting a large resource.
- Scanning a large PowerCenter repository.
- Scanning a large-volume business intelligence report.
- Scanning an Oracle source with large volume public synonym.
- Increasing upgrade throughput when you move to a higher version.
- Profiling millions of unstructured files.

To optimize the performance, you can consider the following recommendations:

- Split a large resource into multiple resources, if possible.
- Increase the disk space by 10 GB for every additional 1 million assets.
- When a single large resource exceeds the **High** scanner memory option limit, you can increase the memory for the Catalog Service using the custom properties for low, medium, or high based on your requirement. For example, in a high load type deployment, you can increase the memory using the **LdmCustomOptions.scanner.memory.high** value as necessary in the **Informatica Administrator** > **Manage** > **Services and Nodes** > **Catalog Service** > **Custom Properties** section.

  For example, when you want to run complex mappings based on large resources, large reports, or large unstructured files, or if you want to include all the schemas in a resource scan, then increase the custom property value.

  Alternatively, you can use the *-Dscanner* JVM option to increase the memory for the specific resource when you configure the resource.
- When you want to run a profile on millions of unstructured files, increase the node process heap size to 2 GB.

  To increase the node process heap size, perform the following steps:

  1. Open the **infaservice.sh** file located in the `<INFA_HOME>/tomcat/bin/` directory.
  2. Increase the value of the *INFA_JAVA_OPTS* parameter from **INFA_JAVA_OPTS="-Xmx1024m"** to **INFA_JAVA_OPTS="-Xmx2048m"**.
  3. Save the file.
  4. Restart the Informatica domain.
- When the number of files and tables in a resource exceed 100,000, increase the Model Repository Service heap size to 4 GB.

  To increase the heap size, navigate to **Informatica Adminstrator** > **Services and Nodes** > **Model Repository Service** > **Properties** > **Advanced Properties** section and set **Maximum Heap Size** to **2048m**.

## *Metadata Extraction Scanner Memory Parameters*

Depending on the size of source data for a resource, you can use one of the following parameters to configure the memory requirements for the scanner to extract metadata:

| Parameter | Memory Option* | Supported Object Count for Default Memory[1] | Default Memory[2] |
|---|---|---|---|
| LdmCustomOptions.scanner.memory.low | low | Up to 1000 tables<br>Up to 2000 files<br>Up to 100 mappings/reports<br>Up to 100,000 objects<br>Up to a maximum file size of 5 MB for unstructured files | 2 GB |
| LdmCustomOptions.scanner.memory.medium | medium | Up to 10,000 tables<br>Up to 20,000 files<br>Up to 1000 mappings/reports<br>Up to 1M objects<br>Up to a maximum file size of 10 MB for unstructured files | 4 GB |
| LdmCustomOptions.scanner.memory.high | high | Up to 100,000 tables<br>Up to 1,000,000 files<br>Up to 20,000 Mappings/Reports<br>Up to 10M objects<br>Up to a maximum file size of 100 MB for unstructured files | 12 GB |

* Indicates the memory allocated to scanner process heap size.

[1] Depends on the number of columns in the table, file size, file type, complexity of the mapping, size of the report, and object content.

[2] Indicates the default values configured for the scanner based on the data size.

Increase the value of the **LdmCustomOptions.scanner.memory.high** custom property as necessary or configure the *-Dscanner.memory.high=<Memory in MB> JVM* option when one of the following conditions is true:

- You want to run complex mappings based on large resources, large reports, or large unstructured files.
- You want to include all the schemas in a resource scan.
- A single large resource exceeds the High scanner memory option limit.

To increase the value of the **LdmCustomOptions.scanner.memory.high** custom property, perform the following steps:

1. In Informatica Administrator, navigate to the **Manage** > **Services and Nodes** > **Catalog Service** > **Custom Properties** section.
2. Edit the property, enter the required value, and save the property.

To configure the *-Dscanner.memory.high=<Memory in MB> JVM* option, perform the following steps:

1. In Catalog Administrator, edit the resource.
2. In the **Metadata Load Settings** > **Advanced Properties** section, configure the following options:
   - Memory. Choose **High**.

- JVM Options. Enter **-Dscanner.memory.high=<Memory in MB>**.

3. Save and run the resource.

**Note:**

- You must select the number of concurrent scanners based on the memory type and the available resources in the cluster.
- You must increase the Informatica PowerCenter scanner memory based on the complexity of the mappings.

## Database Management Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from database resources based on the resource memory configured:

### Low

The scanner memory configured is 2 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Oracle | Table | 1K | 100K |
| MS SQL | Table | 1K | 100K |
| SAP HANA | Table | 5K | 50K |
| Teradata | Table or View | 1K | 10K |
| MS SQL Stored Procedure | Stored Procedure | 5K | 5K |

### Medium

The scanner memory configured is 4 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Oracle | Table | 10K | 1 million |
| MS SQL | Table | 10K | 1 million |
| SAP HANA | Table | 10K | 100K |
| Teradata | Table or View | 10K | 1 million |
| MS SQL Stored Procedure | Stored Procedure | 10K | 10K |

### High

The scanner memory configured is 12 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Oracle | Table | 100K | 5 million |
| MS SQL | Table | 100K | 5 million |
| SAP HANA | Table | 20K | 200K |
| Teradata | Table or View | 20K | 5 million |
| MS SQL Stored Procedure | Stored Procedure | 20K | 20K |

## Advanced Scanners Database Stored Procedure Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from Advanced Scanners database stored procedure resources based on the resource memory configured:

### Low

The scanner memory configured is 2 GB.

| Resource Type | Data Type | Advanced Scanners Asset Count | Enterprise Data Catalog Asset Count |
|---|---|---|---|
| Oracle | Stored procedure | 500 | 15K |
| MS SQL | Stored procedure or view | 500 | 25K |
| Teradata | Stored procedure | 500 | 15K |

### Medium

The scanner memory configured is 4 GB.

| Resource Type | Data Type | Advanced Scanners Asset Count | Enterprise Data Catalog Asset Count |
|---|---|---|---|
| Oracle | Stored procedure | 1K | 50K |
| MS SQL | Stored procedure or view | 1K | 50K |
| Teradata | Stored procedure | 1K | 50K |

### High

The scanner memory configured is 12 GB.

| Resource Type | Data Type | Advanced Scanners Asset Count | Enterprise Data Catalog Asset Count |
|---|---|---|---|
| Oracle | Stored procedure | 5K | 125K |
| MS SQL | Stored procedure or view | 2K | 100K |
| Teradata | Stored procedure | 5K | 125K |

## Data Integration Resources Memory Recommendations

The following load type tables list the maximum number of assets that you can extract from data integration resources based on the resource memory configured:

### Low

The scanner memory configured is 2 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| PowerCenter | Mapping | 1K | 100K |
| IBM DataStage | Mapping | 1K | 10K |
| Oracle Data Integrator | Mapping | 10 | 1K |
| IICS | Mapping | 5K | 100K |
| Big Data Management | Mapping | 1K | 1 million |
| Big Data Streaming | Mapping | 500 | 100K |
| SSIS | Mapping or File | 100 | 100K |
| Custom Lineage | Links | 100K | - |
| Database Scripts | Scripts | 500 | 500 |

### Medium

The scanner memory configured is 4 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| PowerCenter | Mapping | 10K | 1 million |
| IBM DataStage | Mapping | 5K | 50K |
| Oracle Data Integrator | Mapping | 100 | 10K |
| IICS | Mapping | 10K | 1 million |

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Big Data Management | Mapping | 5K | 5 million |
| Big Data Streaming | Mapping | 1K | 250K |
| SSIS | Mapping or File | 500 | 2 million |
| Custom Lineage | Links | 250K | - |
| Database Scripts | Scripts | 1K | 1K |

### High

The scanner memory configured is 12 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| PowerCenter | Mapping | 25K | 5 million |
| IBM DataStage | Mapping | 10K | 100K |
| Oracle Data Integrator | Mapping | 250 | 50K |
| IICS | Mapping | 20K | 2 million |
| Big Data Management | Mapping | 10K | 10 million |
| Big Data Streaming | Mapping | 2K | 500K |
| SSIS | Mapping or File | 1K | 5 million |
| Custom Lineage | Links | 1 million | - |
| Database Scripts | Scripts | 5K | 5K |

## Advanced Scanners Data Integration Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from the Advanced Scanners data integration resources based on the resource memory configured:

### Low

The scanner memory configured is 2 GB.

| Resource Type | Data Type | Advanced Scanners Asset Count | Enterprise Data Catalog Asset Count |
|---|---|---|---|
| SSIS | Mapping | 250 | 12.5K |
| Talend Data Integration | Mapping | 500 | 10K |

| Resource Type | Data Type | Advanced Scanners Asset Count | Enterprise Data Catalog Asset Count |
|---|---|---|---|
| Oracle Data Integrator | Mapping | 1K | 10K |
| IBM DataStage | Mapping | 250 | 25K |

### Medium

The scanner memory configured is 4 GB.

| Resource Type | Data Type | Advanced Scanners Asset Count | Enterprise Data Catalog Asset Count |
|---|---|---|---|
| SSIS | Mapping | 500 | 25K |
| Talend Data Integration | Mapping | 1K | 20K |
| Oracle Data Integrator | Mapping | 2K | 20K |
| IBM DataStage | Mapping | 500 | 50K |

### High

The scanner memory configured is 12 GB.

| Resource Type | Data Type | Advanced Scanners Asset Count | Enterprise Data Catalog Asset Count |
|---|---|---|---|
| SSIS | Mapping | 1K | 50K |
| Talend Data Integration | Mapping | 2K | 5K |
| Oracle Data Integrator | Mapping | 5K | 50K |
| IBM DataStage | Mapping | 2.5K | 250K |

## Business Intelligence Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from business intelligence resources based on the resource memory configured:

### Low

The scanner memory configured is 2 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| MicroStrategy | Reports | 500 | 10K |
| Qlik Sense | Reports | 500K | 500K |
| SAP BusinessObjects | Reports | 100 | 25K |

### Medium

The scanner memory configured is 4 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| MicroStrategy | Reports | 1K | 50K |
| Qlik Sense | Reports | 1 million | 1 million |
| SAP BusinessObjects | Reports | 1K | 500K |

### High

The scanner memory configured is 12 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| MicroStrategy | Reports | 5K | 100K |
| Qlik Sense | Reports | 5 million | 5 million |
| SAP BusinessObjects | Reports | 5K | 1 million |

## Custom Scanner Resource Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from a custom scanner resource based on the resource memory configured:

### Low

The scanner memory configured is 2 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Custom scanner | Objects | 5K | 5K |

### Medium

The scanner memory configured is 4 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Custom scanner | Objects | 10K | 10K |

### High

The scanner memory configured is 12 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Custom scanner | Objects | 20K | 20K |

## No SQL Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from No SQL resources based on the resource memory configured:

### Low

The scanner memory configured is 2 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| MongoDB | Collections | 1K | 100K |
| Apache Cassandra | Table | 1K | 100K |

### Medium

The scanner memory configured is 4 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| MongoDB | Collections | 5K | 1 million |
| Apache Cassandra | Table | 10K | 1 million |

### High

The scanner memory configured is 12 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| MongoDB | Collections | 10K | 5 million |
| Apache Cassandra | Table | 100K | 5 million |

## Cloud Data Warehouse Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from cloud data warehouse resources based on the resource memory configured:

### Low

The scanner memory configured is 2 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Google BigQuery | Table | 1K | 1 million |
| Snowflake | Table | 1K | 100K |
| Amazon Redshift | Table | 1K | 100K |
| ADLS | File | 10K | 10K |

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Amazon S3 | File | 10K | 10K |
| Google Cloud Storage | File | 10K | 10K |
| AWS Glue | Object | 100K | 1 million |

### Medium

The scanner memory configured is 4 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Google BigQuery | Table | 5K | 5 million |
| Snowflake | Table | 5K | 500K |
| Amazon Redshift | Table | 5K | 500K |
| ADLS | File | 50K | 10K |
| Amazon S3 | File | 50K | 10K |
| Google Cloud Storage | File | 50K | 10K |
| AWS Glue | Object | 250K | 5 million |

### High

The scanner memory configured is 12 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Google BigQuery | Table | 10K | 10 million |
| Snowflake | Table | 10K | 1 million |
| Amazon Redshift | Table | 10K | 1 million |
| ADLS | File | 100K | 10K |
| Amazon S3 | File | 100K | 10K |
| Google Cloud Storage | File | 100K | 10K |
| AWS Glue | Object | 500K | 2 million |

## Streaming Resource Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from streaming resource based on the resource memory configured:

### Low

The scanner memory configured is 2 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Apache Kafka | Topics | 1K | 25K |

### Medium

The scanner memory configured is 4 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Apache Kafka | Topics | 5K | 50K |

### High

The scanner memory configured is 12 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Apache Kafka | Topics | 10K | 100K |

## Data Engineering Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from data engineering resources based on the resource memory configured:

### Low

The scanner memory configured is 2 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Apache Atlas | Links | 5K | 5K |
| Cloudera Navigator | Entities | 5K | 5K |
| Apache Hive | Tables | 1K | 100K |
| HDFS | Files | 1K | 10K |

### Medium

The scanner memory configured is 4 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Apache Atlas | Links | 10K | 10K |
| Cloudera Navigator | Entities | 10K | 10K |

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Apache Hive | Tables | 10K | 1 million |
| HDFS | Files | 10K | 1 million |

### High

The scanner memory configured is 12 GB.

| Resource Type | Data Type | Data Type Count | Maximum Asset Count |
|---|---|---|---|
| Apache Atlas | Links | 20K | 20K |
| Cloudera Navigator | Entities | 20K | 20K |
| Apache Hive | Tables | 50K | 5 million |
| HDFS | Files | 100K | 5 million |

## Advanced Scanners Data Analytics and Reporting Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from Advanced Scanners data analytics and reporting resources based on the resource memory configured:

### Low

The scanner memory configured is 2 GB.

| Resource Type | Data Type | Advanced Scanners Asset Count | Enterprise Data Catalog Asset Count |
|---|---|---|---|
| SAS | Scripts | 250 | 250K |
| SSAS | Analysis files | 1K | 200K |
| SSRS | Reports | 1K | 15K |
| Snowflake | Tables | 1K | 200K |

### Medium

The scanner memory configured is 4 GB.

| Resource Type | Data Type | Advanced Scanners Asset Count | Enterprise Data Catalog Asset Count |
|---|---|---|---|
| SAS | Scripts | 500 | 500K |
| SSAS | Analysis files | 2K | 400K |
| SSRS | Reports | 2K | 30K |
| Snowflake | Tables | 2K | 400K |

### High

The scanner memory configured is 12 GB.

| Resource Type | Data Type | Advanced Scanners Asset Count | Enterprise Data Catalog Asset Count |
|---|---|---|---|
| SAS | Scripts | 1K | 1 million |
| SSAS | Analysis files | 5K | 1 million |
| SSRS | Reports | 5K | 100K |
| Snowflake | Tables | 10K | 2 million |

## Advanced Scanners Legacy Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from Advanced Scanners legacy resources based on the resource memory configured:

### Low

The scanner memory configured is 2 GB.

| Resource Type | Data Type | Advanced Scanners Asset Count | Enterprise Data Catalog Asset Count |
|---|---|---|---|
| COBOL scripts | Scripts | 250 | 2.5K |
| COBOL programs using JCL scripts | Scripts | 250 | 10K |

### Medium

The scanner memory configured is 4 GB.

| Resource Type | Data Type | Advanced Scanners Asset Count | Enterprise Data Catalog Asset Count |
|---|---|---|---|
| COBOL scripts | Scripts | 500 | 5K |
| COBOL programs using JCL scripts | Scripts | 500 | 25K |

### High

The scanner memory configured is 12 GB.

| Resource Type | Data Type | Advanced Scanners Asset Count | Enterprise Data Catalog Asset Count |
|---|---|---|---|
| COBOL scripts | Scripts | 1K | 10K |
| COBOL programs using JCL scripts | Scripts | 1K | 100K |

## Tuning Configuration Parameters for Resources

You can configure parameters to optimize the performance of resources.

### Cloud Resources

You can tune the performance parameters for cloud resources such as Azure Data Lake Storage and Google Cloud Storage resources.

To increase the source metadata extraction throughput, tune the **-DfileProcessorCount** JVM option. By default, this option is set to **2** for all file-based resources.

You can configure the **-DfileProcessorCount** parameter based on the number of cores required for a node in the cluster as shown in the following table:

| Number Of Cores Required for a Node In The Cluster | JVM Option and Value |
|---|---|
| 8 | DfileProcessorCount=8 |
| 16 | DfileProcessorCount=16 |

To add, edit, or delete a JVM option, navigate to the **Metadata Load Settings** tab of the resource in Catalog Administrator.

## Tuning Performance for Similarity

Enterprise Data Catalog stores the similarity and value frequency details in the PostgreSQL database.

Similarity discovery between resources is calculated based on column name, data pattern in the column, and the unique values in the columns. As similarity is a resource-intensive process, it is recommended that you plan for a disk space based on the data volume. The minimum recommendation for disk space is 200 GB to support a similarity resource group with two resources and a total of approximately 10,000 columns in total.

The following table lists the heap size recommendations for similarity resource groups:

| Load Type | Heap Memory | Total Number of Colums per Similarity Resource Group |
|---|---|---|
| Low | 2 GB | Up to 10K columns. |
| Medium | 4 GB | Up to 100K columns. |

You can tune the computation throughput by adding additional hardware for the PostgreSQL database. You can configure the additional hardware using the `LdmCustomOptions.scanner.simDiscoveryScanner.partitionPairJobCount` Catalog Service custom property. Default value for the property is 5 for a PostgreSQL node that includes 8 CPU cores and 16 GB of memory. To increase the value by 5 for the custom property, you must add 4 CPU cores and 4 GB of memory.

The recommended number of resources in a similarity resource group is 4. Based on the number of columns in a resource group and the available hardware, you can use the `LdmCustomOptions.scanner.simDiscoveryScanner.maxPermittedResourceCount` Catalog Service custom property to specify the number of resources in a similarity resource group.

## Enterprise Data Catalog Agent

You can use Enterprise Data Catalog Agent to extract metadata from data sources that run on Microsoft Windows.

The following table lists the recommended system requirements to deploy Enterprise Data Catalog Agent:

| Component | Minimum System Requirements |
|---|---|
| CPU Cores | 8 |
| Memory | 16 GB |

For more information, see Appendix D: Enterprise Data Catalog Agent in the *Catalog Administrator Guide*.

## Log Files

To monitor the performance and to identify issues, you can view the log files. Log files are generated at every stage of metadata extraction and ingestion.

You can view the following log files for Enterprise Data Catalog:

**Accessing the Log files from the UI**

You can access the log files from the following Enterprise Data Catalog URL: `http://<host name>:<Catalog Service port>/access/ui/index`

**Catalog Service Log File**

Navigate to the `<$INFA_HOME>/logs/<node_name>/services/CatalogService/<Catalog_service_name>/LDM.log` directory to view the Catalog Service log file.

**Scanner Log File**

In Catalog Administrator, you can configure the scanner log level for a resource and MITI log level for data sources that support MITI. You can set the log level as NONE, FATAL, ERROR, WARNING, STATUS, INFO, or DEBUG.

To configure the log level, perform the following steps:

- To configure the scanner log level, navigate to the resource and set the following JVM option in the **Metadata Load Settings** > **Advanced Properties** section:
  `-Dscannerloglevel=<scanner_log_level>`

  For example, `-Dscannerloglevel=INFO`
- To configure the MITI log level, navigate to the resource and set the following JVM option in the **Metadata Load Settings** > **Advanced Properties** section:
  `-DagentLogLevel=<scanner_log_level>`

  For example, `-DagentLogLevel=INFO`.

**Note:** By default, the scanner log level is INFO and log level for data sources supported by MITI is DEBUG. It is recommended to reduce the log level when the log file size increases in GB during a single large resource run.

In Catalog Administrator, you can view the Log URL for resources in the **Monitoring** > **Tasks** section.

**MongoDB Log Files**

You can access the MongoDB log files from the directory that you configured for the **MongoDB Log Directory** property when you created the Informatica Cluster Service. Default is `$clusterCustomDir/mongo/log`.

**Apache ZooKeeper Log Files**

You can access the Apache ZooKeeper log files from the directory that you configured for the **ZooKeeper Data Directory** property when you created the Informatica Cluster Service. Default is `$clusterCustomDir/zk/data`.

**Apache Solr Log Files**

You can access the Apache Solr log files from the directory that you configured for the **Solr Data Directory** property when you created the Informatica Cluster Service. Default is `$clusterCustomDir/solr/data`.

**Nomad Aggregation Log Files**

You can view the Nomad aggregation log files for high volume load types using the `nomad alloc logs -f <Task_ID> > /path` command. You can get the <Task_ID> from the Nomad UI.

**PostgreSQL Log Files**

You can access the PostgreSQL log files from the directory that you configured for the **PostgreSQL DB Log Directory** property when you created the Informatica Cluster Service. Default is `$clusterCustomDir/postgres/log`.

**Data Integration Service Log File**

To view the log file, navigate to the `<$INFA_HOME>/logs/<node_name>/services/DataIntegrationService/disLogs/dis_<dis_ service_name>_<node_name>.log.0` directory.

**Mapping Log File**

To view the log file, navigate to the `<$INFA_HOME>/logs/<node_name>/services/DataIntegrationService/disLogs/profiling/<logs>` directory.

**XMI File Logs**

You can use the following flag when you configure a resource to retain the XMI log files: `-DpersistXmiLocation=<path in the cluster>`

# Performance Tuning Parameters for Profiling

Tuning profiling performance involves configuring the Data Integration Service parameters, Data Integration Service concurrency parameters, and profiling warehouse parameters.

## *Tuning Data Integration Service Parameters for Profiling*

To optimize data profiling, you can configure Data Integration Service parameters, Data Integration Service profiling tuning parameters, and Data Integration Service grid configuration. You can also tune unique key inference performance parameters.

**Note:** For optimal performance, configure Data Integration Service on a standalone node or multiple nodes with horizontal scaling instead of a grid.

You need to configure the Temporary Directories and Maximum Execution Pool Size parameters for the Data Integration Service. You can configure parameters, such as Reserved Profile Threads that apply to the Profiling Service Module. Before you use the parameter recommendations, verify that you have identified a node or grid and the requirement is to configure the node or grid optimally to run profiles.

The following configuration parameters have an impact on different components of the profiling and discovery installation. You can increase or decrease the values for these parameters based on the performance requirements:

| Parameter | Description |
|---|---|
| Maximum Execution Pool Size | The maximum number of requests that the Data Integration Service can run concurrently. Requests include data previews, mappings, and profiling jobs. This parameter has an impact on the Data Integration Service. |
| Maximum Profile Execution Pool Size | The total number threads to run profiles. This parameter has an impact on the Profiling Service Module. |
| Temporary Directories | Location of temporary directories for the Data Integration Service process on the node. This parameter has an impact on the Data Integration Service machine. |

**Note:** For more information about Data Integration Service parameters, see the *Profiling and Discovery Sizing Guidelines* article.

## Data Integration Service Profiling Tuning Parameters

The following table lists the values for the Data Integration Service profiling tuning parameters:

| Tuning Parameter | Value for 100,000 or lesser Rows with Row Sampling Enabled | Value for 100,000 or more Rows with Row Sampling Disabled | Description |
|---|---|---|---|
| Maximum Concurrent Columns | 100 | 5 | Maximum number of columns merged in single mapping. |
| Maximum Column Heap Size | 512 | 512 | Maximum cache size allocated for a mapping. |
| AdvancedProfilingServiceOptions.ColumnsThreshold | 100 | Default | Maximum number of columns merged for a mapping. Applicable only for column profiling. |
| AdvancedProfilingServiceOptions.DomainMappingColumnThreshold | 100 | 50 | Maximum number of columns merged for a mapping. Applicable only for data domain discovery. |
| AdvancedProfilingServiceOptions.MaxDataDomains | 150 | Default | Maximum number of data domains combined for a data domain discovery mapping. |
| ExecutionContextOptions.Optimizer.MaxMappingPorts | -1 | Default | Controls platform optimization during mapping execution. |
| ExecutionContextOptions.Optimizer.PushdownType | none | Default | Controls pushdown optimization during mapping execution. |

| Tuning Parameter | Value for 100,000 or lesser Rows with Row Sampling Enabled | Value for 100,000 or more Rows with Row Sampling Disabled | Description |
|---|---|---|---|
| ExecutionContextOptions.Optimizer.ExtractSourceConstraints | false | Default | Controls extracting constraint details for a relational source during mapping execution. |
| ExecutionContextOptions.JVMOption1 | 1024M | 1024M | Controls mapping process heap size. |
| ExecutionOptions.MaxProcessLifeTime | 1800000 | Default | Controls mapping process life time. |
| ExecutionContextOptions.Optimizer.NativeExpressionTx | false | Default | Controls SISR optimization during mapping execution. |

## Data Integration Service Node Scaling for Profiling

The execution time and throughput for data discovery varies based on the following parameters. It also depends on the hardware configuration of the Data Integration Service node.

- Volume of data in the tables and files
- Number of rows
- Number of unique or duplicate data sets
- Number of columns
- Column length
- Sampling options
- Number of data domains

You can choose a configuration from the following options to scale the Data Integration Service node for profiling:

- **Option 1 — Vertical hardware scaling with a standalone node**
  - This option is recommended if you have a single profiling resource.
  - The throughput for a single profiling resource can be increased by vertically scaling the node hardware configuration.
  - To achieve vertical scaling, you must calculate the Data Integration Service concurrency parameters for profiling and increase the number of CPU cores and system memory on the node. For information on how to calculate the Data Integration Service concurrency parameters, refer to the "Data Integration Service Concurrency Parameters" on page 35.
- **Option 2 — Horizontal hardware scaling with multiple nodes**
  - This option is recommended if you have multiple profiling resources.
  - The throughput for multiple profiling resources can be increased by horizontally scaling the node hardware configuration. This configuration is used to increase the throughput across resources.
  - Create multiple nodes with dedicated profiling warehouse schema that can be selected at the resource level under the domain connection settings.

- **Option 3 — Grid configuration**
  - This option is used to increase the throughput for a single profiling resource by horizontal scaling.
  - Using this configuration, you can improve the data discovery throughput by up to 1.4x by increasing the total hardware capacity by 2x across multiple grid nodes.
  - The generation of mappings and compilation is performed on the master node. The execution takes place on both the master and the secondary nodes. Therefore, the hardware utilization on the master node is always higher than the secondary nodes.
  - The minimum hardware requirement recommended for a node used in a grid configuration is 32 CPU cores and 64 GB memory.
  - For information on how to configure multiple nodes in a grid configuration, refer to the "Data Integration Service Grid Configuration for Enterprise Data Catalog" on page 34 section.

## Data Integration Service Grid Configuration for Enterprise Data Catalog

In Enterprise Data Catalog, to use the Data Integration Service on a grid, you need to configure a shared directory with read and write access on all the nodes in the grid.

To configure the *<INFA_HOME>/<profile_temp_folder>* shared directory on all the nodes, perform the following steps:

1. Log in to Informatica Administrator.
2. Navigate to **Manage** > **Services and Nodes** > **Domain Navigator**.
3. Select the grid in the **Domain Navigator**.
4. In the **Properties** > **Custom Properties** section, click the **Edit** icon.
5. In the **Edit Custom Properties** dialog box, click **New**.
6. In the **New Custom Property** dialog box, enter the following details:
   a. In the **Name** field, enter **AdvancedProfilingServiceOptions.CustomFlag8**.
   b. In the **Value** field, enter the temporary folder. For example, `/data2/profile_temp`.
7. Recycle the Data Integration Service.

In Enterprise Data Catalog, you can use the shared directory to perform the following tasks:

- To stage XML or JSON schemas on a Data Integration Service node
- To stage unstructured files and results on a Data Integration Service node
- To stage similarity results on a Data Integration Service node

## Tuning Performance for Unique Key Inference Process

The unique key inference throughput and execution time are exponential and they depend on the following factors:

- Schema size
- Data set volume
- Profiling sample size
- Profiling concurrency

To tune unique key inference performance, configure the **ExecutionContextOptions.JVMoption1 = -Xmx4096M** JVM option in the **Informatica Administrator** > **Manage** > **Services and Nodes** > **Data Integration Service** > **Custom Properties** section.

## *Data Integration Service Concurrency Parameters*

To increase the profiling performance throughput, you can tune the following Data Integration Service concurrency parameters:

- Maximum On-Demand Execution Pool Size
- Maximum Profile Execution Pool Size
- Maximum Concurrent Profile Jobs
- Maximum Concurrent Profile Threads
- AdvancedProfilingServiceOptions.FileConnectionLimit. This parameter applies to SAP, XML, and JSON sources.
- AdvancedProfilingServiceOptions.HiveConnectionLimit. This parameter applies to Hive sources.
- Maximum # of Connections for source connection

You can configure the concurrency parameters in one of the following ways:

- In the Administrator tool, navigate to the Data Integration Service properties to configure the concurrency parameters.
- In the command prompt, navigate to the `<INFA_HOME>/isp/bin/plugins/ps/concurrencyParameters/` location.
  The *ConcurrencyParameters.properties* file in this location contains the default values of the Data Integration Service concurrency parameters for each load type. Run the *setConcurrencyParameters* script with the load type option to update the parameter values simultaneously for Data Integration Service.

### Calculate Concurrency Parameter Values

You can calculate the value for concurrency parameters based on the number of cores. It is recommended that you multiply the number of cores with 0.6 and round off the value to the nearest whole value. For example, if the number of cores configured for the Data Integration Service is 16, then the concurrency parameter value is calculated as follows: 16*0.6=9.6. The concurrency parameter value is rounded off to the nearest whole value, namely, 10.

The following table describes how to calculate the Data Integration Service concurrency parameters for profiling based on the number of cores:

| Concurrency Parameter | Value* | Description |
|---|---|---|
| Maximum On-Demand Execution Pool Size | Number of cores x 0.6 | Maximum parallel mappings run by the Data Integration Service that includes all types of mappings. |
| Maximum Profile Execution Pool Size | Number of cores x 0.6 | Maximum parallel profiling mappings run by the Data Integration Service. |
| Maximum Concurrent Profile Jobs | Number of cores x 0.6 | Maximum parallel tables or files processed. |
| Maximum Concurrent Profile Threads | Number of cores x 0.6 | Maximum parallel mappings run by the Data Integration Service for File sources. |
| AdvancedProfilingServiceOptions.FileConnectionLimit (For CSV, XML, and JSON metadata sources) | Number of cores x 0.6 | Maximum parallel profiling mappings run by the Data Integration Service for File sources. |

| Concurrency Parameter | Value* | Description |
|---|---|---|
| AdvancedProfilingServiceOptions.HiveConnectionLimit (For Hive metadata sources) | Number of cores x 0.6 | Maximum parallel profiling mappings run by the Data Integration Service for Hive sources. |
| Maximum # of Connections | Number of cores x 0.6 | Total number of database connections for each resource. |
| * Indicates the value of the parameter if the number of rows is less than or greater than 100, 000 rows with sampling of rows enabled or disabled. | | |

## setConcurrencyParameters Script

You can run the script to update all the parameter values in the *ConcurrencyParameters.properties* file simultaneously. After you run the script, the script stops the Data Integration Service, updates the parameter values in the Data Integration Service, and then enables the Data Integration Service.

The *Maximum # Connection Pool Size* parameter is configured at the data source level. Therefore, you need to update this parameter in the Administrator tool for relational sources.

The setConcurrencyParameters script uses the following syntax:

```
setConcurrencyParameters.sh
[small/medium/large]
[DomainName]
[DISName]
[Username]
[Password]
```

The following table describes the setConcurrencyParameters script arguments:

| Argument | Description |
|---|---|
| small/medium/large | Required. Enter small, medium, or large. |
| DomainName | Required. Enter the domain name. |
| DISName | Required. Enter the Data Integration Service name. |
| Username | Required. Enter the user name that the Data Integration Service uses to access the Model Repository Service. |
| Password | Required. Enter the password for the Model repository user. |

**Usage**

```
./setConcurrencyParameters.sh [small/medium/large] [DomainName] [DISName] [Username]
[Password]
```

**Example**

```
./setConcurrencyParameters.sh small domain DIS Administrator Administrator
```

**Default Parameter Values for Small, Medium, and Large Options in the Script**

| Parameter | Small | Medium | Large |
|---|---|---|---|
| Data Integration Service server hardware | CPU cores is16.<br>Memory is 32 GB. | CPU cores is 32.<br>Memory is 64 GB. | CPU cores is 64.<br>Memory is 128 GB. |
| Maximum On-Demand Execution Pool Size | 10 | 20 | 40 |
| Maximum Profile Execution Pool Size | 10 | 20 | 40 |
| Maximum Concurrent Profile Jobs | 10 | 20 | 40 |
| Maximum Concurrent Profile Threads | 10 | 20 | 40 |
| AdvancedProfilingServiceOptions.FileConnectionLimit<br>(For SAP, XML, JSON sources) | 10 | 20 | 40 |
| AdvancedProfilingServiceOptions.HiveConnectionLimit<br>(For Hive sources) | 10 | 20 | 40 |
| Maximum # of Connections for source connection | 10 | 20 | 40 |

## *Tuning Profiling Warehouse Parameters*

The profiling warehouse stores profiling results. More than one Profiling Service Module may point to the same profiling warehouse. The main resource for the profiling warehouse is disk space.

The following table lists the parameters that you need to tune to improve the performance of the profiling warehouse based on the size of data:

| Parameter | Low | Medium | High | Description |
|---|---|---|---|---|
| Tablespace | 50 GB | 200 GB | 500 GB | The maximum amount of disk space required to temporarily store the profiling results before these results get stored in the Enterprise Data Catalog. |
| CPU | 4 | 8 | 16 | The number of cores required by the profiling warehouse |

**Note:** For more information about the profiling warehouse guidelines, see the Profiling and Discovery Sizing Guidelines article.

## *Tuning Performance for Profiling on Spark Engine*

To increase the profiling throughput for high volume sources, run the profiles on the Spark engine in the Hadoop environment.

To optimize profiling performance, run the following resources on the Spark engine:

- HDFS sources
- Hive sources
- Partitioned parquet files on Amazon S3 and Microsoft Azure Data Lake Store Gen2.

To tune the profiling throughput, configure the following parameters:

- Increase the value of **spark.yarn.executor.memoryOverhead** from 384 to 2048.
- Configure the **spark.dynamicAllocation.maxExecutors** parameter to 5 for every 3 nodes. For example, you can configure **spark.dynamicAllocation.maxExecutors** to 10 for 6 nodes.

To configure the parameters, navigate to the **Informatica Administrator** > **Manage** > **Connections** > **Hadoop CCO connection** > **Spark Configuration** > **Advanced Properties** section.

The minimum system requirement for each node in the Spark cluster is 32 CPU cores and 128 GB of memory. The following table describes the Service Level Agreement (SLA) to process metadata extraction and profiling:

| Load Type | Number of Cluster Nodes | Hive Schema Size | Number of Tables per Schema | Number of Columns per Table | Recommended Sampling | Number of Data Domains | SLA |
|---|---|---|---|---|---|---|---|
| Medium | 3 | Upto 10 | 100 | 100 | 1 million | 10 | 48 - 72 Hours |
| High | 6 | Upto 100 | 200 | 100 | 10 million | 10 | 1 - 4 Weeks |

| Load Type | Number of Cluster Nodes | Hive Schema Size | Number of Tables per Schema | Number of Columns per Table | Recommended Sampling | Number of Data Domains | SLA |
|---|---|---|---|---|---|---|---|
| Very High | 12 | Upto 100 | 1000 | 100 | 10 million | 10 | 4 - 10 Weeks |

**Note:**
- The SLA is based on Hive table execution performance on an on-premise HDP/CDH cluster.
- The recommended mode for profiling is random percentage sampling. The maximum recommended sampling for the Spark engine is 10 million. You can extrapolate the sampling based on the actual row count.
- The profiling execution time and throughput will vary based on whether the data source is cloud based or resides on a local file system. It also depends on the following factors:
  - Data source type
  - Volume of content in the tables or files
  - Number of tables, rows, and columns
  - Column length
  - File type
  - Average and maximum file size
  - Number of unique and duplicate datasets
  - Sampling mode
  - Number of data domains
  - Source system hardware configuration
  - Disk latency
  - Network latency between source systems
- The data domain discovery execution time is linearly proportional to the total number of data domains selected for profiling.
- The cluster performance will vary based on the underlying hardware configuration of the cluster nodes and the cluster type such as EMR/HDP/CDP.

For information about best practices and to tune the Spark engine, see the Tuning and Sizing Guidelines for Informatica® Data Engineering Integration 10.4.0 and Performance Tuning for the Spark Engine articles.

## *Tuning Performance for Profiling on Blaze Engine*

To increase the profiling throughput for high volume sources, you can run the profiles on the Blaze engine in the Hadoop environment for relational sources that exceed 100 million rows.

The following table lists the custom properties for the Data Integration Service and the values you need to configure to run the profiles on the Blaze engine:

| Parameter | Value |
|---|---|
| ExecutionContextOptions.Blaze.AllowSortedPartitionMergeTx | false |
| ExecutionContextOptions.GridExecutor.EnableMapSideAgg | true |
| ExecutionContextOptions.RAPartitioning.AutoPartitionResult | true |
| AdvancedProfilingServiceOptions.BlazeBatchSize | 1 |
| AdvancedProfilingServiceOptions.DomainMappingColumnThreshold | 1 |
| AdvancedProfilingServiceOptions.ColumnsThreshold | 1 |

**Note:** When you run profiles on the Blaze engine, the profiling service stages the source data in the cluster nodes which uses higher disk space than the source size.

For information about best practices and to tune the Blaze engine, see the *Performance Tuning and Sizing Guidelines for Informatica Big Data Management* and *Best Practices for Big Data Management Blaze Engine* articles.

## Memory Sizing to Run Profiles on Large Unstructured Files

You can run a profile on unstructured files, such as PDF, Microsoft Excel, and Microsoft Word files.

### Memory Requirements to Run Profiles on Large PDF Data Sources

Large unstructured files in PDF format require more memory during profiling.

The following table lists the memory that you can configure for the *ExecutionContextOptions.JVMOption1* parameter based on the PDF file size:

| PDF File Size | Memory (in MB) |
|---|---|
| Less than or equal to 25 MB | 1024M |
| Less than or equal to 50 MB | 2048M |
| Less than or equal to 100 MB | 5120M |

### Memory Requirements to Run Profiles on Large Microsoft Excel or Microsoft Word Files

The following table lists the memory that you can configure for the *ExecutionContextOptions.JVMOption1* parameter based on the Microsoft Excel or Microsoft Word file size:

| File Size | Memory (in MB) |
|---|---|
| Less than or equal to 5 MB | 1024M |
| Less than or equal to 25 MB | 2048M |
| Less than or equal to 100 MB | 7168M |

To tune the *ExecutionContextOptions.JVMOption1* parameter, navigate to the **Manage** > **Services and Nodes** > **Data Integration Service** > **Custom Properties** section in Informatica Administrator.

## Tuning Resource Configuration Parameters for Profiling

### Snowflake Resource

To optimize the profiling performance for Snowflake high volume resources, configure the following custom properties in Informatica Administrator:

- **AdvancedProfilingServiceOptions.ColumnsThreshold**. Set the value to **10**.
- **ExecutionContextOptions.JVMoption1**. Set the value to **-Xmx2048M**.

To configure these parameters, navigate to the **Services and Nodes** > **Data Integration Service** > **Custom Properties** section.

## Microsoft Azure Data Lake Store Gen2 Resource

You can configure the following parameters to optimize the profiling performance for Microsoft Azure Data Lake Store Gen2 resources:

**Heap Size for pmdtmsvc2 Process**

To set the maximum heap size for pmdtmsvc2 process used for profiling Microsoft Azure Data Lake Store Gen2 high volume resources in the native runtime environment, set the following custom property :

`ExecutionContextOptions.JVMoption1` custom property to -Xmx4096M value.

The pmdtmsvc2 process memory requirement might increase for a large file. When the configured heap size does not meet the file processing memory requirement, a heap size error appears and the mapping is marked as failed.

To configure the custom property, navigate to the **Informatica Administrator** > **Services and Nodes** > **Data Integration Service** > **Custom Properties** section.

**Concurrent Threads**

You can set concurrent threads to download the files from Microsoft Azure Data Lake Store Gen2 source. By default, this option is set to 10. It is recommended to configure this value in the range of current number of cores to twice the number of CPU cores. For example, on a 16 CPU cores server, you can configure this parameter from 16 through 32.

You can configure the following option at resource level to override the default value:

`-Dconnector.access.concurrent.thread=<value>`

To configure this JVM option at resource level in Catalog Administrator, open the resource and set the JVM option on the **Metadata Load Settings** tab.

**Block Size**

You can set the block size to download the files in chunks using multiple threads. By default, the option is set to 8MB.

You can configure this option at resource level to override the default value:

`-Dconnector.access.block.sizeInMB=<value>`

It is recommended that you consider the pmdtmsvc2 process heap size and concurrent threads to configure this option to an optimum value to avoid redundant network calls.

To configure this JVM option at resource level in Catalog Administrator, open the resource and set the JVM option on the **Metadata Load Settings** tab.

**Heap Size**

Configure the Data Integration Service heap size to 10GB in Informatica Administrator before you run the data domain discovery profiles in Catalog Administrator. You can increase this value if the data domains exceed 100.

## Cassandra Resource

You can configure the following parameters to optimize the profiling performance for Cassandra resources:

**Heap Size for pmdtmsvc2 Process**

Configure the **ExecutionContextOptions.JVMoption1** custom property to -Xmx2048M to optimize the profiling performance for Cassandra high volume resources. This parameter sets the maximum heap size for pmdtmsvc2 process used for profiling.

To configure these parameters, navigate to **Informatica Administrator** > **Services and Nodes** > **Data Integration Service** > **Properties** > **Custom Properties** section.

**Heap Size**

Configure the Data Integration Service heap size to 10 GB when you run data domain discovery on Cassandra resources. Increase the heap size if the number of data domains exceed 100.

## *Profiling Recommendations for a Databricks Cluster*

Verify the following recommendations to optimally run profiling on a Databricks cluster:

- The Databricks cluster and source data ,must reside in the same geographical region to reduce network latency.
- For better throughput and performance, you can enable autoscaling.

### Hardware Recommendations for Databricks Runtime 5.5 LTS

Verify the following hardware recommendations:

- You can configure the worker or driver type node as Standard_DS3_v3 that includes 128.0 GB of memory, 32 CPU cores, and 6 DBU.
- You can configure a minimum of 1 worker node and a maximum of 8 worker nodes with autoscaling enabled.

# Catalog Export and Migration Recommendations

As part of upgrade to version 10.5, you must export, back up, and migrate the catalog data. You must also back up the Model Repository and upgrade the Model Repository Service with the backed up content as part of the upgrade process. Verify the recommendations when you upgrade to version 10.5.
Detailed instructions about content backup, export, and migration are included in *Upgrading from Versions 10.4.0 and Later*

The following sections list the recommendations for upgrade:

### Model Repository Backup and Upgrade Recommendations

See the following table for the heap size recommendation based on the size of the Model Repository backup file:

| Model Repository Backup File Size | Recommended Heap Size |
|---|---|
| 200 MB | 4 GB |
| 200 MB-2 GB | 8 GB |
| 2 GB-4 GB | 12 GB |

You can configure the MRS heap size from Informatica Administrator. Access the Model Repository Service **Properties** > **Advanced Properties** section from Informatica Administrator and set the recommended heap size based on the file size.

## Catalog Data Backup and Export Recommendations

See the following table for the heap size recommendation to back up and export catalog data:

| Asset Count | Recommended Heap Size | Approximate Time for Data Backup in Hours |
|---|---|---|
| 2 million | 10 GB | 0.5 |
| 10 million | 10 GB | 1 |
| 20 million | 10 GB | 2 |
| 50 million | 15 GB | 5 |
| 100 million | 20 GB | 10 |

The expected throughput for backup and export is 10 million assets per hour.

For a medium load type, the default memory configuration is sufficient for the export of 20 million assets with the Informatica domain node configured with 32 GB of memory. If the Informatica domain node is shared by multiple applications, such as Data Privacy Management, Enterprise Data Catalog, and Enterprise Data Preparation, you can restrict the maximum heap memory used by the migration process. You can use the `java-Xmx<memory>m-DskipReadOnlyCheck=true -jar export.jar export` option with the export utility as shown in the following sample to restrict the maximum heap memory: `java-Xmx10240m-DskipReadOnlyCheck=true -jar export.jar export`.

For a high load type, the default memory configuration is sufficient for the export of 50 million assets with the Informatica domain node configured with 64 GB of memory. If the Informatica domain node is shared by multiple applications, such as Data Privacy Management, Enterprise Data Catalog, and Enterprise Data Preparation, you can restrict the maximum heap memory used by the migration process. You can use the `java-Xmx<memory>m-DskipReadOnlyCheck=true -jar export.jar export` option with the export utility as shown in the following sample to restrict the maximum heap memory: `java-Xmx15360m-DskipReadOnlyCheck=true -jar export.jar export`.

## Catalog Data Migration Recommendations

See the following table for the heap size recommendation and the estimated time to migrate data based on the number of assets in the catalog:

| Asset Count | Recommended Heap Size | Approximate Time for Data Migration in Hours |
|---|---|---|
| 2 million | 6 GB | 1.3 |
| 10 million | 10 GB | 3 |
| 20 million | 10 GB | 6 |
| 50 million | 15 GB | 15 |
| 100 million | 20 GB | 30 |

The expected throughput for migration is 2-4 million assets per hour.

For a medium load type, the default memory configuration is sufficient for the migration of 20 million assets with the Informatica domain node configured with 32 GB of memory. If the Informatica domain node is shared by multiple applications, such as Data Privacy Management, Enterprise Data Catalog, and Enterprise Data Preparation, you can restrict the maximum heap memory used by the migration process. You can set the

`xdocs.content.systemProperties=` variable to `-Xmx10G` in the `$INFA_HOME/services/CatalogService/Binaries/MigrationModuleConfigurations.properties` file to restrict the maximum heap memory.

For a high load type, the default memory configuration is sufficient for the migration of 50 million assets with the Informatica domain node configured with 64 GB of memory. If the Informatica domain node is shared by multiple applications, such as Data Privacy Management, Enterprise Data Catalog, and Enterprise Data Preparation, you can restrict the maximum heap memory used by the migration process. You can set the `xdocs.content.systemProperties=` variable to `-Xmx15G` in the `$INFA_HOME/services/CatalogService/Binaries/MigrationModuleConfigurations.properties` file to restrict the maximum heap memory.

### *Catalog Re-index Estimates*

After you migrate the catalog data, and you observe missing data links, you can re-index the catalog to resolve the issue.
See the following table for an approximate re-indexing time based on the volume of data in the catalog:

| Asset Count in Catalog Data | Approximate Re-indexing Time in Hours |
|---|---|
| 2 million | 0.30 |
| 10 million | 1 |
| 20 million | 2 |
| 50 million | 5 |
| 100 million | 10 |

If the catalog data includes a high volume of shared links, it is recommended to increase the publish task memory from the default value of 4 GB to 8 GB. You can use the `LdmCustomOptions.ingest.publish.task.memory.mb` Catalog Service custom property to increase the default publish task memory value.

## Enterprise Data Catalog Performance Tuning Parameters

You can tune the Enterprise Data Catalog performance by configuring the custom properties for the Catalog Service.

The following table lists the custom properties and parameters that you can use to tune Enterprise Data Catalog performance:

| Custom Properties and Parameters | Type | Default Value | Description |
|---|---|---|---|
| LdmCustomOptions.orchestration.nomad.exec.additional.jvm.flags | JVM configuration | You can use the following JVM options to debug:<br>- -XX:+HeapDumpOnOutOfMemoryError<br>- -XX:HeapDumpPath=/data1/HEAPDUMP/<br>- -XX:+PrintGCTimeStamps<br>- -XX:+PrintGCDetails<br>- -XX:+PrintGCApplicationStoppedTime<br>- -XX:+PrintGCApplicationConcurrentTime<br>- -XX:+PrintHeapAtGC<br>- -Xloggc:/data1/Analysis/gc.lo<br>**Note:** You must separate the JVM options with a white space character. | Flag to debug JVM performance. |
| - Catalog Service configuration. LdmCustomOptions.scanner.memory.low<br>- Scanner configuration. -Dscanner.memory.low | Memory configuration | Default is 2048 MB. | You can use this parameter for specific scenarios where the default value needs to be increased. Alternatively, you can increase the resource memory from low > medium > high. |
| - Catalog Service configuration. LdmCustomOptions.scanner.memory.medium<br>- Scanner configuration. -Dscanner.memory.medium | Memory configuration | Default is 4096 MB. | You can use this parameter for specific scenarios where the default value needs to be increased. Alternatively, you can increase the resource memory from low > medium > high. |
| - Catalog Service configuration. LdmCustomOptions.scanner.memory.high<br>- Scanner configuration. -Dscanner.memory.high | Memory configuration | Default is 12288 MB. | You can use this parameter for specific scenarios where the default value needs to be increased. Alternatively, you can increase the resource memory from low > medium > high. |

| Custom Properties and Parameters | Type | Default Value | Description |
|---|---|---|---|
| - Catalog Service configuration. LdmCustomOptions.ingest.resolve.task.memory.mb<br>- Scanner configuration. - Dingest.resolve.task.memory.mb | Memory configuration | Default is 4096 MB. | You can use this parameter to improve ingestion performance for high volume sources. Verify that adequate amount of free memory is available on all cluster nodes. |
| - Catalog Service configuration. LdmCustomOptions.ingest.publish.task.memory.mb<br>- Scanner configuration. - Dingest.publish.task.memory.mb | Memory configuration | Default is 4096 MB. | You can use this parameter to improve publishing performance for high volume sources. Verify that adequate amount of free memory is available on all cluster nodes. |
| - Catalog Service configuration. LdmCustomOptions.ingest.assign.task.memory.mb<br>- Scanner configuration. - Dingest.assign.task.memory.mb | Memory configuration | Default is 4096 MB. | You can use this parameter to improve connection assignment performance for high volume sources. Verify that adequate amount of free memory is available on all cluster nodes. |
| - Catalog Service configuration. LdmCustomOptions. ingest.purge.task.memory.mb<br>- Scanner configuration. - Dingest.purge.task.memory.mb | Memory configuration | Default is 4096 MB. | You can use this parameter to improve purge performance for high volume sources. Verify that adequate amount of free memory is available on all cluster nodes. |
| - Catalog Service configuration. LdmCustomOptions.ingest.publish.object.batch.size<br>- Scanner configuration. - Dingest.publish.object.batch.size | Tune publish jobs batch size tuning. | Default is 1000. | You can use this parameter to improve publish batch size performance for high volume sources. **Attention:** Use the parameter with caution as increasing the batch size might result in CPU throttling that might impact overall performance and product stability. |

| Custom Properties and Parameters | Type | Default Value | Description |
|---|---|---|---|
| LdmCustomOptions.orchestration.exec.engine.concurrent.jobs | Tune concurrent jobs per node. | Default is 1 per node. | You can use this parameter to increase the number of concurrent jobs running on a node.<br>**Attention:** The CPU and memory requirements per node increases if you increase the value for this parameter, which might lead to stability and performance issues with the MongoDB database. The memory that you configure for the node must be based on the MongoDB WiredTiger cache memory associated with the MongoDB license. |
| -DfileProcessorCount | Resource configuration to tune concurrent file processors. | Default is 2. | You can use this parameter to increase the number of concurrent file processors.<br>**Attention:** Increasing the value might lead to CPU throttling issues and thread blocks. |
| -Dconnector.access.concurrent.thread | Resource configuration to tune concurrent threads to download files from a data lake. | Default is 10. | You can use this parameter to increase the number of concurrent threads to download files from a data lake.<br>**Attention:** Increasing the value might lead to CPU throttling issues and thread blocks. |
| -Dconnector.access.block.sizeInMB | Resource configuration to tune concurrent Threads to download file from Data lake | Default is 8 MB. | You can use this parameter to increase the number of concurrent threads to download files from a data lake in a high volume scenario.<br>To prevent redundant network calls, it is recommended to tune this value based on the values specified for the pmdtmsvc2 process heap size and concurrent threads. |

| Custom Properties and Parameters | Type | Default Value | Description |
|---|---|---|---|
| -Dsolrconfig.updatehandler.autocommit.maxdocs | Resource configuration | Default is 1000. | Parameter to govern maximum number of Apache Solr documents stored in-memory before an auto commit is performed. |
| -Dscannerloglevel - | Resource log files | You must add the parameter to the custom properties in the `EDCConfiguration.properties` file under the following directory: `<INFA_HOME>/services/CatalogService/Binaries/`<br>Set the number of log files for the resources using the orchestration.job.log.max.file = <value> parameter.<br>Set the log file size for the resources using the orchestration.job.log.max.file.size = <value> parameter.<br>To validate the details, you must verify the Nomad configuration for each job from the log configuration: Job > TaskGroup > Task > logConfig | Parameter for resource debugging.<br>**Attention:** Use the parameter as required as it might impact disk utilization for high volume resources. |
| -DskipTestConnection | Resource configuration | - | You can use this parameter to skip testing the connection properties configured for a resource. |
| -DskipExtraction | Resource configuration | - | You can use this parameter to skip extracting metadata for a resource. |

| Custom Properties and Parameters | Type | Default Value | Description |
|---|---|---|---|
| LdmCustomOptions.scanner.simDiscoveryScanner.maxPermittedResourceCount | Configure resources in Similarity discovery group. | Default is 4. | Parameter to increase the number of resources in the similarity resource group.<br>**Attention:** Similarity Computation is a resource-intensive process and might require additional disk space based on the volume of data.<br>For all high volume load types, it is recommended to increase the number of resources per resource group only if additional hardware is available.<br>**Note:** You must restart the Catalog Service if you configure this parameter. |
| LdmCustomOptions.scanner.simDiscoveryScanner.partitionPairJobCount | Configure Similarity computation and throughput | Default is 5. | You can increase the similarity throughput by adding PostgreSQL database hardware. You can then tune the throughput using the property. The default value of 5 represents 8 CPU cores and 16 GB of memory for a PostgreSQL node. To increase the count by 5, you must add an addition 4 CPU cores and 4 GB of memory for the PostgreSQL node.<br>**Note:** You must restart the Catalog Service if you configure this parameter. |
| ICS Mongo DB Service Options.Mongo Custom Options.MemoryLimitPercentage | Configure MongoDB WiredTiger cache memory | Default is 50 | Parameter to control the memory allocated for MongoDB WiredTiger cache. For better stability, it is recommended that you set the value of this parameter to 40.<br>**Note:** This parameter is only applicable for 10.5.0.1 and later versions. |

If you encounter `OutOfMemory` errors for PowerCenter resources enabled for reference resources, you can resolve the issue by setting the `LdmCustomOptions.ingest.use.new.lineage.publish.imp` custom property to `false`.

# Performance Tuning Parameters for Data Asset Analytics

Performance tuning involves configuring parameters for the Data Asset Analytics repository database.

## *Tuning PostgreSQL Database Parameters for Data Asset Analytics*

If you configured PostgreSQL as the repository database for Data Asset Analytics, perform the following steps to tune Data Asset Analytics performance:

- If you have 30 million or lesser number of assets in the catalog, configure the PostgreSQL parameters as shown in the following table:

| Parameter | Value |
|---|---|
| work_mem | 1 GB |
| shared_buffers | 16 GB |
| effective_cache_size | 15 GB |
| effective_io_concurrency | 200 |
| min_wal_size | 2 GB |
| max_wal_size | 8 GB |
| max_worker_processes | 16 |
| max_parallel_workers_per_gather | 4 |
| max_parallel_workers | 16 |
| max_parallel_maintenance_workers | 4 |
| random_page_cost | 1.1 |
| max_locks_per_transaction | 1024 |

  **Note:** If the catalog includes more than 30 million assets, you can tune the parameters based on your requirement.

- Add the `pg_hint_plan` extension in the PostgreSQL database to use existing `hints` in SQL statements and generate the optimal execution plan.
- Add the `BatchMechanism=COPY` parameter in the JDBC connection string as shown in the following sample:
  `jdbc:Informatica:postgresql://<host name>:<port number>;DatabaseName=<DB Name>;BatchMechanism=COPY`

## *Tuning Oracle Database Parameters for Data Asset Analytics*

The machine that hosts the Oracle database must meet the following minimum recommended system requirements:

| Number of Assets | Component | Minimum Requirement |
|---|---|---|
| Up to 20 million | CPU Cores | 4 |
| | Memory | 24 GB |
| 20 – 50 million | CPU Cores | 8 |
| | Memory | 32 GB |
| 50 – 100 million | CPU Cores | 16 |
| | Memory | 64 GB |

- If you have 20 million or lesser number of assets in the catalog, to tune Data Asset Analytics performance, configure the database parameters as shown in the following table:

| Parameter | Value |
|---|---|
| sga_target | 8 GB |
| pga_aggregate_target | 4 GB |
| pga_aggregate_limit | 6 GB |

- If you have 20-50 million assets in the catalog, to tune Data Asset Analytics performance, configure the database parameters as shown in the following table:

| Parameter | Value |
|---|---|
| sga_target | 12 GB |
| pga_aggregate_target | 6 GB |
| pga_aggregate_limit | 10 GB |

- If you have 50-100 million assets in the catalog, to tune Data Asset Analytics performance, configure the database parameters as shown in the following table:

| Parameter | Value |
|---|---|
| sga_target | 20 GB |
| pga_aggregate_target | 12 GB |
| pga_aggregate_limit | 20 GB |

Configure additional database parameters as shown in the following table:

| Parameter | Value |
|---|---|
| filesystemio_options | NONE or ASYNC |
| processes | 1000 |
| open_cursors | 1000 |
| optimizer_capture_sql_plan_baselines | False |
| optimizer_use_sql_plan_baselines | False |

**Note:**

- The network latency between the machine on which Data Asset Analytics is installed and the database must be in the range of 20-30 milliseconds.
- The temporary tablespace value configured for the database must be a minimum of 50 GB.
- It is recommended that you enable periodic collection of table statistics to ensure optimum performance.
- When the number of assets and concurrent users increase, you must scale up the parameter values and hardware configuration based on the resource usage.

## *Tuning Microsoft SQL Server Database Parameters for Data Asset Analytics*

The machine that hosts the SQL Server database must meet the following minimum recommended system requirements:

| Number of Assets | Component | Minimum Requirement |
|---|---|---|
| Up to 5 million | CPU Cores | 8 |
| | Memory | 32 GB |
| 5 – 10 million | CPU Cores | 12 |
| | Memory | 32 GB |
| 10 – 20 million | CPU Cores | 16 |
| | Memory | 32 GB |
| 20 – 30 million | CPU Cores | 24 |
| | Memory | 32 GB |
| 30 – 50 million | CPU Cores | 32 |
| | Memory | 32 GB |

Perform the following steps to tune Data Asset Analytics performance:

1. Set the value of the cost threshold for parallelism parameter to 5.

2. Run the following command to configure the maximum degree of parallelism: `ALTER DATABASE SCOPED CONFIGURATION SET MAXDOP = <Recommended Value>` based on the number of assets shown in the following table:

| Number of Assets | Recommended Value |
|---|---|
| Up to 5 million | 5 |
| 5 – 20 million | 8 |
| 20 – 50 million | 12 |

**Note:**

- The network latency between the machine on which Data Asset Analytics is installed and the database must be less than 60 milliseconds.
- The disk response time for the database machine must be less than 50 milliseconds.
- It is recommended that you enable periodic collection of table statistics to ensure optimum performance.
- When the number of assets and concurrent users increase, you must scale up the parameter values and hardware configuration based on the resource usage.

## *Tuning Microsoft Azure SQL Database Parameters for Data Asset Analytics*

The following table lists the recommended minimum system requirements for the machine that hosts the SQL Server database on Microsoft Azure:

| Number of Assets | Microsoft Azure Service Tier | Minimum Requirement |
|---|---|---|
| Up to 5 million | vCore - General Purpose (Gen5) | vCores – 16, Storage – 5 GB |
| | vCore - Business Critical | vCores – 8, Storage – 5 GB |
| | DTU - Standard | S9: Number of DTUs – 1600, Storage – 5 GB |
| | DTU - Premium | P6: Number of DTUs – 1000, Storage – 5 GB |
| 5 – 10 Million | vCore - General Purpose (Gen5) | vCores – 32, Storage – 10 GB |
| | vCore - Business Critical | vCores – 20, Storage – 10 GB |
| | DTU - Standard | S12: Number of DTUs – 3000, Storage – 10 GB |
| | DTU - Premium | P11: Number of DTUs – 1750, Storage – 10 GB |
| 10 – 20 Million | vCore - General Purpose (Gen5) | vCores – 32, Storage – 20 GB |
| | vCore - Business Critical | vCores – 20, Storage – 20 GB |
| | DTU - Standard | Not Recommended |
| | DTU - Premium | P15: Number of DTUs – 4000, Storage – 20 GB |

| Number of Assets | Microsoft Azure Service Tier | Minimum Requirement |
|---|---|---|
| 20 – 30 Million | vCore - General Purpose (Gen5) | vCores – 40, Storage – 20 GB |
| | vCore - Business Critical | vCores – 32, Storage – 20 GB |
| | DTU - Standard | Not Recommended |
| | DTU - Premium | P15: Number of DTUs – 4000, Storage – 20 GB |

Set the value of the maximum degree of parallelism parameter to a minimum of 12.

Run the following command to configure the maximum degree of parallelism: `ALTER DATABASE SCOPED CONFIGURATION SET MAXDOP = 12`.

**Note:**

- The network latency between the machine on which Data Asset Analytics is installed and the database must be less than 60 milliseconds.
- It is recommended that you enable periodic collection of table statistics to ensure optimum performance.
- When the number of assets and concurrent users increase, you must scale up the parameter values and hardware configuration based on the resource usage.

## *Tuning Amazon Web Services RDS for SQL Server Database Parameters*

The following table lists the recommended minimum system requirements for the machine that hosts the SQL Server database on Amazon RDS:

| Number of Assets | Component | Minimum Requirement |
|---|---|---|
| Up to 10 million | vCPU | 8 |
| | Storage Type | General Purpose SSD |
| | Disk Size | 20 GB |
| 10 – 20 million | vCPU | 16 |
| | Storage Type | Provisioned IOPS |
| | IOPS | 2000 |
| | Disk Size | 100 GB |
| 20 – 30 million | vCPU | 32 |
| | Storage Type | Provisioned IOPS |
| | IOPS | 5000 - 6000 |
| | Disk Size | 200 GB |

Perform the following step to tune Data Asset Analytics performance:

Run the following command to configure the maximum degree of parallelism: `ALTER DATABASE SCOPED CONFIGURATION SET MAXDOP = <Recommended Value>` based on the number of assets shown in the following table:

| Number of Assets | Recommended Value |
|---|---|
| Up to 20 million | 8 |
| 20 – 30 million | 12 |

**Note:**

- The network latency between the machine on which Data Asset Analytics is installed and the database must be less than 60 milliseconds.
- It is recommended that you enable periodic collection of table statistics to ensure optimum performance.
- When the number of assets and concurrent users increase, you must scale up the parameter values and hardware configuration based on the resource usage.

# Author

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